

ACTIVITY 7 — LEARNER WORKSHEET**Pareto, Run Charts, Variation, Yield, DPU, and DPMO**

TYPE	DURATION	DMAIC PHASE	MAPS TO
Core Activity	75 min	ANALYZE	A3, A4

Name	
Date	

What you are doing

Turn collected data into decisions. Apply the Pareto principle to find the vital few categories, use a run chart to see behaviour over time, and calculate the standard Six Sigma process metrics.

What you will produce: A Pareto chart, a run chart with the pattern named, and calculated yield, DPU, DPO, DPMO and sigma level that reconcile with the data.

Tools and techniques: Pareto Chart tool (collaborative), NovaSPC, yield/DPU/DPO/DPMO, sigma level, variation

The scenario

Contoso Service Desk handles employee IT requests. Employees complain that tickets take too long to be assigned, that they must keep chasing for status, and that agents apply different definitions of what counts as "resolved". Handoffs between the front line and the specialist queues are unclear, and although ticket data exists it is inconsistent. You are supporting the improvement team as a Yellow Belt — you do not lead the project.

Two weeks of data have been collected: 400 tickets, of which 96 contained at least one defect, with 120 defects recorded in total across 6 defect opportunities per ticket. Mean assignment time is 55 minutes (median 46) against a 30-minute improvement goal. The CSV files in this activity's data/ folder are that extract.

Files in this activity folder**data/ — the mock data you analyse:**

- assignment-times.csv — 400 rows (date, ticket_id, assignment_time_min)
- daily-summary.csv — 10 rows (date, tickets, mean_assignment_min, median_assignment_min, defects, missed_30min_goal)
- defect-counts.csv — 6 rows (defect_category, count)
- service-desk-tickets.csv — 400 rows (ticket_id, date, ticket_type, channel, assigned_queue, agent, assignment_time_min, met_30min_goal, defect_count, defect_categories, reopened)

templates/ — the worksheets you complete:

- pareto-table.csv (defect_category, count, percent, cumulative_percent)
- process-metrics.csv (metric, formula, your_calculation)

Open these in Excel, Google Sheets or LibreOffice Calc. Work on a copy so the originals stay clean for revision.

Upload data/assignment-times.csv to NovaSPC for the run chart. Worked answers are in solution/pareto-table-answers.csv and solution/process-metrics-answers.csv — attempt the activity before opening them.

Your steps

1. Build the Pareto table: category, count, percentage and cumulative percentage.
2. In your group, open the collaborative Pareto tool — one member creates the session and shares the code, then everyone joins, brainstorms and votes to produce the live Pareto chart.

Tool: <https://alfredang.github.io/paretochart/>

3. Identify the vital few categories driving about 80% of the problem.
4. Export your assignment-time data as CSV and plot a run chart in NovaSPC.

Tool: <https://alfredang.github.io/novaspc/>

5. Interpret the run chart for trend, shift, cluster and oscillation patterns.
6. Calculate yield, DPU, DPO and DPMO from your defect data.
7. Convert DPMO to a sigma level and interpret what it says about the process.
8. Distinguish common-cause from special-cause variation and why the response differs.

Your working space

Complete every field. You may keep this worksheet and refer to it during the open-book assessment.

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Step 3 — Identify the vital few categories driving about 80% of the problem.

Step 4 — Export your assignment-time data as CSV and plot a run chart in NovaSPC.

Step 5 — Interpret the run chart for trend, shift, cluster and oscillation patterns.

Step 6 — Calculate yield, DPU, DPO and DPMO from your defect data.

Step 7 — Convert DPMO to a sigma level and interpret what it says about the process.

Step 8 — Distinguish common-cause from special-cause variation and why the response differs.

Answers to the discussion prompts

1. Which two categories make up the vital few, and what percentage do they carry?

2. What pattern does the run chart show — trend, shift, cluster or oscillation?

3. What can you NOT yet conclude from the Pareto alone?

Reflect & discuss

What would a sigma level of ~3.1 mean if it were your own process?

Where in your work do you react to a single data point instead of the pattern?

Check your work

Your cumulative percentage column reaches 100%, and you can state the sigma level with the DPMO it came from.